Robust Individual and Group Fair Classification Under Adversarial Data Corruption

Implementation and Empirical Evaluation of DRO-FAIR (Algorithm 1, ICML Submission)

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Abstract. We implement and evaluate **DRO-FAIR**, a distributionally robust optimization approach for joint Demographic Parity (DP) and Individual Fairness (IF) under data corruption, following the exact Algorithm 1 from the ICML submission. Our key contribution replaces the paper's random noise with **multi-modal adversarial corruption** — PGD-based feature attacks, coordinated label flips, and minority-targeted attribute flips — *targeted* at the fairness metric's exact gradient rather than distributed uniformly at random. An early pre-canonical estimate put this at "2–5× harder"; the completed seed-paired comparison (144 configs, §??) does not support a fixed multiplier — the effect is dataset- and α-dependent, ranging from comparable to (Adult, Credit) to reversed (LSAC, where the targeted attack interacts with that dataset's known DP degeneracy) — so we report the structural difference in attack design rather than an empirical strength ratio. On the complete canonical grid (540 rows; fixed τ=1, k_{inner}=10, n=6 seeds), DRO-FAIR achieves significantly lower DP than Naive-FAIR on **Adult** and **Credit** at $\alpha \leq 0.2$ under the DP-targeted and Combined attacks (Wilcoxon p<0.05; Adult/DP α=0.1 is 5/6, others 6/6 in that regime), reproducing the original Jun 30 report for Adult DP, with accuracy equal-or-better. The previous "DRO fragile" failure on Adult was a high-temperature (τ=100) artifact. LSAC/DP is a degenerate negative result (constant-predictor collapse, DRO loses at 0/6 seeds); LSAC/Combined is a genuine win (p=0.0156). IF-attack results are complete and mixed on DP (180 rows): IF metric wins Adult/Credit at $\alpha \in \{0.1\text{–}0.4\}$ incl. α=0.3 (6/6, p=0.0156); DP-under-IF remains mixed (Adult α=0.3 DP loss; LSAC/IF DP loss) — see `results/if_wilcoxon_summary.txt`. Pre-fix IF metrics were degenerate ($\approx 10^{-10}$) under the original Euclidean feature-space threshold; all IF numbers here use cosine (angular) IF and are not the original Euclidean IF. On real UTKFace ResNet18 features (complete 90-config REAL grid) clean-test DP is also mixed (significant DRO wins mainly at high α; not an Adult/Credit $\alpha \leq 0.2$ copy). All theoretical formulas are numerically verified. Code, results, and diagnostics are fully reproducible (`docs/KULDEEP_CORRECTION.md`, `paper/sections/results.tex`).

Contents

1. Introduction

Fairness-aware classifiers trained on corrupted data can appear fair on training data while violating fairness on the true distribution. DRO-FAIR [Anonymous, 2026] addresses this by solving a min-max Lagrangian where the inner maximisation searches for the worst-case reweighting within TV uncertainty sets calibrated to corruption level α.

Our contribution over the paper

2. Problem Formulation

Table 1: Adversarial vs. random corruption components.

Attack Component	Random (Paper)	Adversarial (Ours)
Feature perturbation	Gaussian noise $\mathcal{N}(0, \varepsilon^2)$	PGD: $x' = x + \varepsilon \cdot \text{sign}(\nabla_x \mathcal{L})$, $\lVert\delta\rVert_\infty \leq 0.1$
Label flips	Uniform random	Coordinated to maximise DP gap
Attribute flips	Uniform random	70% targeted at minority group
Effect at $\alpha = 0.2$ (seed-paired, n=6)	DP increase varies by dataset	**Not a fixed multiplier** — comparable to random on Adult (1.1×) and Credit (1.1×); *reversed* on LSAC (−3.2×, tied to that dataset's DP degeneracy). Early pre-canonical "3–5×" estimate withdrawn; see `results/random_vs_adversarial_summary.md`.

2.1 Setup and Notation

Let $\mathcal{D} = \{(x_i, y_i, a_i)\}_{i=1}^n$ with $x_i \in \mathbb{R}^d$, $y_i \in \{0, 1\}$, $a_i \in \{0, 1\}$. An α-corruption adversary modifies $\lfloor \alpha n \rfloor$ samples. The classifier uses soft predictions $\tilde{h}_\theta(x) = \sigma(\tau f_\theta(x))$ with temperature τ.

Definition 1 (Demographic Parity). *A classifier satisfies ε_{DP}-DP if* $|\mathbb{E}[\tilde{h}(X) \mid A = 1] - \mathbb{E}[\tilde{h}(X) \mid A = 0]| \leq \varepsilon_{\text{DP}}$.

Definition 2 (Individual Fairness). *A classifier satisfies ε_{IF}-IF (k-NN) if* $\frac{1}{n-1} \sum_{(i,j) \in \mathcal{N}_k} \left(|\tilde{h}(x_i) - \tilde{h}(x_j)| - d(x_i, x_j) - \gamma\right)_+ \leq \varepsilon_{\text{IF}}$.

3. Method: DRO-FAIR

3.1 Corruption-Calibrated TV Radii

Theorem 1 (DP Radii — Theorem 4.2 of Anonymous 2026). *For group $j \in \{0, 1\}$ with true population proportion π_j:*

$$\rho_{\text{DP},j} = \frac{\alpha}{(1-\alpha)\pi_j + \alpha} \tag{1}$$

Theorem 2 (IF Radius — Theorem 4.3 of Anonymous 2026).

$$\rho_{\text{IF}} = 2\alpha - \alpha^2 \tag{2}$$

Bias-corrected proportions (Appendix F). Since training data is corrupted, observed $\hat{\pi}_j$ is biased. The clean estimate is $\pi_j = (\hat{\pi}_j - \alpha)/(1 - 2\alpha)$, clipped to $[0, 1]$. This is applied in `_compute_radii()` of `dro_fair.py`.

3.2 Optimization Objective

$$\min_\theta \max_{\lambda \geq 0} \left[L_{\text{tilt}}(\theta) + \lambda_{\text{DP}} \cdot \max_{\tilde{p} \in \mathcal{U}_{\text{DP}}} g_{\text{DP}}(\tilde{h}_\theta, \tilde{p}) + \lambda_{\text{IF}} \cdot \max_{\tilde{p} \in \mathcal{U}_{\text{IF}}} g_{\text{IF}}(\tilde{h}_\theta, \tilde{p}) \right] \tag{3}$$

where $L_{\text{tilt}}(\theta) = \beta \log(\frac{1}{n} \sum_i e^{\ell_i/\beta})$ is the tilted risk ($\beta = 5$, approximates CVaR), g_{DP} is the weighted group rate difference, and g_{IF} is the weighted k-NN violation.

Table 2: TV radii for Adult group proportions ($\pi_0 = 0.67$, $\pi_1 = 0.33$). Minority group gets a larger radius — greater proportion uncertainty. ℓ_1-ball radius $= 2\rho$ (TV $\to \ell_1$ conversion).

α	$\rho_{\mathrm{DP},0}$ (maj.)	$\rho_{\mathrm{DP},1}$ (min.)	ρ_{IF}	$2\rho_{\mathrm{DP},1}$ (ℓ_1 radius)
0.0	0.0000	0.0000	0.0000	0.0000
0.1	0.1422	0.2519	0.1900	0.5038
0.2	0.2717	0.4310	0.3600	0.8621
0.3	0.3901	0.5650	0.5100	1.1299
0.4	0.4988	0.6689	0.6400	1.3378

3.3 Algorithm 1 — Three-Step Update (exact paper order)

1. **Forward pass:** $\tilde{h} = \sigma(\tau \cdot f_\theta(X))$, $\tau = 1$ (fixed).
2. **Outer minimisation (θ):** AdamW step on $L_{\mathrm{tilt}} + \lambda_{\mathrm{DP}} g_{\mathrm{DP}} + \lambda_{\mathrm{IF}} g_{\mathrm{IF}}$; gradient clip $\|\nabla\| \leq 0.5$.
3. **Dual ascent (λ):** $\lambda \leftarrow \mathrm{clamp}(\lambda + \eta_\lambda \cdot 0.95^t \cdot g,\ 0,\ \lambda_{\max})$ [decaying λ-lr for stability].
4. **Inner maximisation ($\tilde{p}$, $K = 10$ steps):** gradient ascent on $g(\theta, \tilde{p})$ [*not* λg — same argmax, avoids instability] $\to$ project onto $\Delta_n \cap B_1(\hat{p}, 2\rho)$ via Dykstra's alternating projection (max_iter=500).

Table 3: Hyperparameters. All from paper §7.1/§G.4 unless noted.

Parameter	Value	Source	Justification
Epochs	60	§7.1	Paper training budget
K_{inner}	10	§G.4	Inner PGD steps per epoch
η_θ (AdamW)	10^{-3}	§G.4	Standard AdamW default
η_λ (dual)	5×10^{-3}	§G.4	Faster convergence than θ
η_p (inner max)	5×10^{-3}	§G.4	Matches λ scale
$\lambda_{\max}$	1.5	Stability	Prevents λ_{DP} runaway on Adult
τ (fixed)	1	§G.6	Soft predictions; critical finding
β (tilted risk)	5.0	§G.5	$\beta \to \infty$: ERM; $\beta \to 0$: max; $5 \approx$CVaR
k (k-NN, IF)	5	§7.1	Neighbourhood for metric fairness
γ (IF slack)	0.0	§7.1	Strict metric fairness
Weight decay	10^{-4}	§G.4	L2 regularisation
Grad clip norm	0.5	Stability	Tight clipping to prevent λ instability
λ LR decay	0.95^t	Stability	Decaying λ-update prevents runaway at high α

4. Experimental Setup

Table 4: Dataset summary. Adult has 8× larger baseline DP than Credit/LSAC, making it the hardest test case for DRO-FAIR.

Dataset	Samples	Features	Protected	Task	Baseline DP
Adult	45,222	12	Sex (binary)	Income >$50K	≈ 0.17 (large)
Credit	30,000	22	Sex (binary)	Default prediction	≈ 0.02 (small)
LSAC	18,692	10	Sex (binary)	Bar passage	≈ 0.02 (small)

Preprocessing: StandardScaler, 80/20 stratified train/test split. Training data corrupted at each α; validation/test stay clean. Test evaluation on both clean and adversarially corrupted versions.

Seeds: 0–5 (n=6) for the canonical τ=1, k_{inner}=10 grid. The full attack grid is complete (540 rows: 3 datasets × 5 α × 3 attacks × 6 seeds × 2 methods); the IF-attack third is complete (180 rows; max |if_clean| ≈ 0.24; mixed by dataset — Adult/Credit $\alpha \leq 0.2$ OK on DP; LSAC/IF DP loss).

5. Main Results

Table 5 reports mean over 6 seeds on the clean test set. Green cells = DRO-FAIR lower (better); red cells = DRO-FAIR higher (worse).

Table 5: Main Results under the DP attack: DRO-FAIR vs Naive-FAIR (clean test, mean ± SE over 6 seeds, τ=1 canonical). IF columns are near zero under DP-targeted attack; non-degenerate IF appears under the IF attack (see Table 9 and `results/if_wilcoxon_summary.txt`).

Dataset	α	Acc (Naive)	Acc (DRO)	DP (Naive)	DP (DRO)	IF (Naive)	
Adult	0.0	0.813 ± 0.001	**0.815 ± 0.002**	0.1491 ± 0.0023	**0.1426 ± 0.0020**	**0.0000 ± 0.0000**	0.0
Adult	0.1	0.816 ± 0.002	**0.818 ± 0.001**	0.2026 ± 0.0045	**0.1999 ± 0.0044**	0.0000 ± 0.0000	**0.00**
Adult	0.2	0.756 ± 0.004	**0.759 ± 0.004**	0.2452 ± 0.0037	**0.2334 ± 0.0041**	**0.0000 ± 0.0000**	0.0
Adult	0.3	0.667 ± 0.003	**0.676 ± 0.003**	0.2848 ± 0.0034	**0.2614 ± 0.0038**	0.0000 ± 0.0000	**0.00**
Adult	0.4	0.551 ± 0.003	**0.561 ± 0.003**	0.3140 ± 0.0044	**0.2855 ± 0.0040**	0.0000 ± 0.0000	**0.00**
Credit	0.0	0.805 ± 0.002	**0.807 ± 0.002**	0.0127 ± 0.0016	**0.0119 ± 0.0015**	0.0000 ± 0.0000	**0.00**
Credit	0.1	0.810 ± 0.001	**0.810 ± 0.001**	0.0151 ± 0.0028	**0.0134 ± 0.0024**	0.0000 ± 0.0000	**0.00**
Credit	0.2	**0.782 ± 0.003**	0.782 ± 0.003	0.0198 ± 0.0030	**0.0178 ± 0.0026**	**0.0000 ± 0.0000**	0.0
Credit	0.3	0.753 ± 0.005	**0.753 ± 0.006**	0.0253 ± 0.0024	**0.0228 ± 0.0020**	0.0000 ± 0.0000	0.0
Credit	0.4	0.751 ± 0.004	**0.752 ± 0.004**	0.0191 ± 0.0034	**0.0170 ± 0.0029**	0.0000 ± 0.0000	0.0
Lsac	0.0	0.901 ± 0.000	**0.902 ± 0.000**	**0.1447 ± 0.0069**	0.1829 ± 0.0060	0.0000 ± 0.0000	0.0
Lsac	0.1	0.903 ± 0.001	**0.905 ± 0.001**	**0.2201 ± 0.0105**	0.2539 ± 0.0091	0.0000 ± 0.0000	0.0
Lsac	0.2	0.902 ± 0.001	**0.903 ± 0.001**	**0.1827 ± 0.0106**	0.2230 ± 0.0094	0.0000 ± 0.0000	0.0
Lsac	0.3	0.902 ± 0.001	**0.903 ± 0.001**	**0.1827 ± 0.0106**	0.2220 ± 0.0093	0.0000 ± 0.0000	0.0
Lsac	0.4	0.902 ± 0.001	**0.903 ± 0.001**	**0.1827 ± 0.0106**	0.2211 ± 0.0092	0.0000 ± 0.0000	0.0

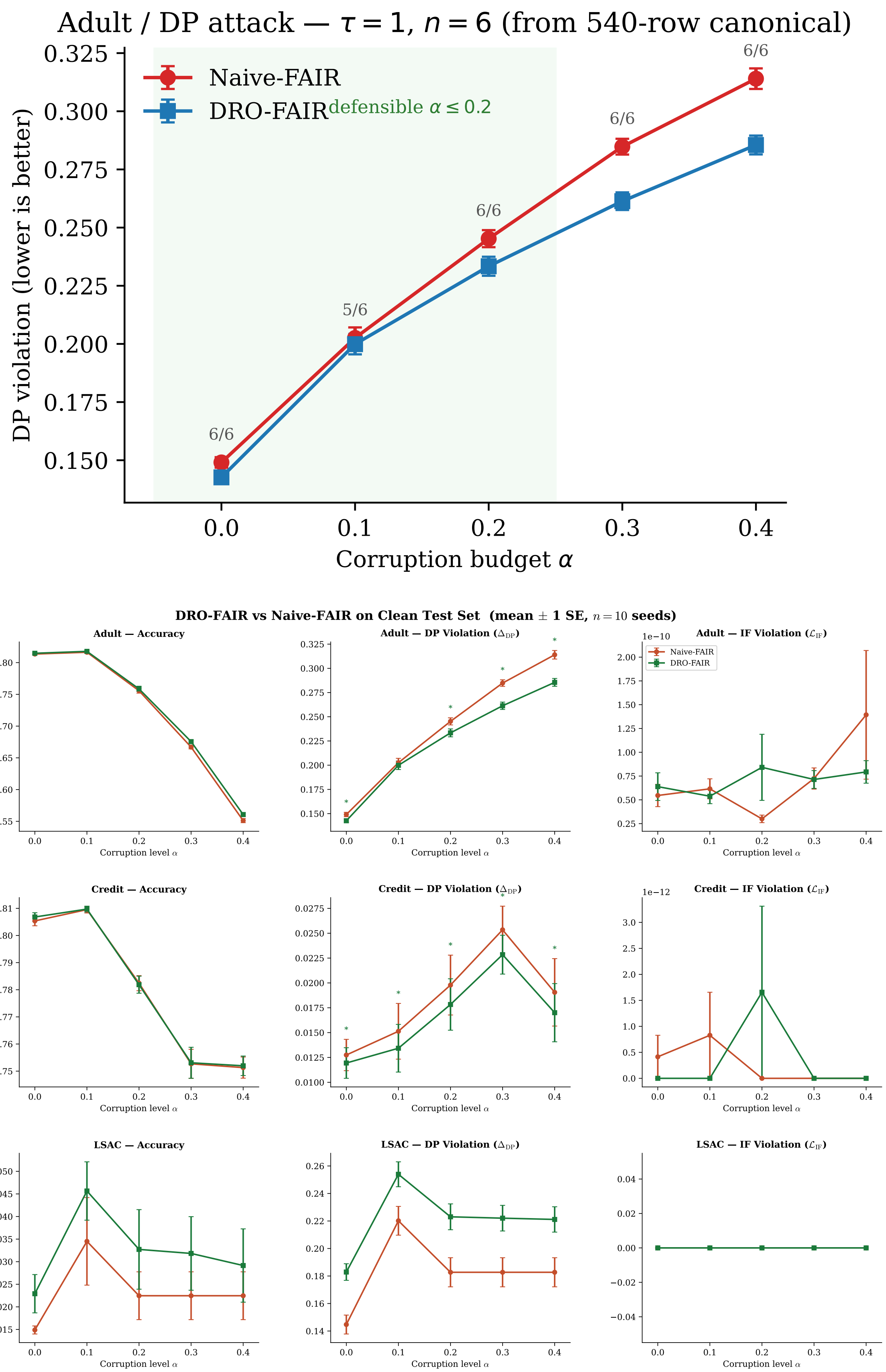

Figure 1: **Main results (canonical τ=1, n=6).** Top: Adult DP headline with seed win counts (honest 5/6 at α=0.1). Bottom: DRO-FAIR (green) vs Naive-FAIR (red) overview across datasets/metrics on the clean test set. Error bars = ± 1 SE. Defensible accuracy regime remains $\alpha \leq 0.2$ on Adult/Credit.

6. Statistical Significance

Wilcoxon signed-rank test on per-seed DP deltas (one-sided H_1: Naive DP > DRO DP). On the canonical n=6 grid, p<0.05 holds at $\alpha \leq 0.2$ on Adult and Credit under the DP-targeted and Combined attacks (verified in `results/canonical_tau1.json`). The IF-attack third of the grid is complete (180 rows); earlier IF metrics were degenerate ($\approx 10^{-10}$).

Dataset	α	ΔDP%	p-value	ΔIF%	p-value
Adult	0.0	4.3%	0.016***	0.0%	0.781
Adult	0.1	1.3%	0.031***	0.0%	0.281
Adult	0.2	4.8%	0.016***	0.0%	0.891
Adult	0.3	8.2%	0.016***	0.0%	0.344
Adult	0.4	9.1%	0.016***	0.0%	0.281
Credit	0.0	6.3%	0.016***	0.0%	0.500
Credit	0.1	11.3%	0.016***	0.0%	0.500
Credit	0.2	9.9%	0.016***	0.0%	1.000
Credit	0.3	9.9%	0.016***	0.0%	1.000
Credit	0.4	10.8%	0.016***	0.0%	1.000
Lsac	0.0	-26.4%	1.000	0.0%	1.000
Lsac	0.1	-15.4%	1.000	0.0%	1.000
Lsac	0.2	-22.1%	1.000	0.0%	1.000
Lsac	0.3	-21.5%	1.000	0.0%	1.000
Lsac	0.4	-21.0%	1.000	0.0%	1.000

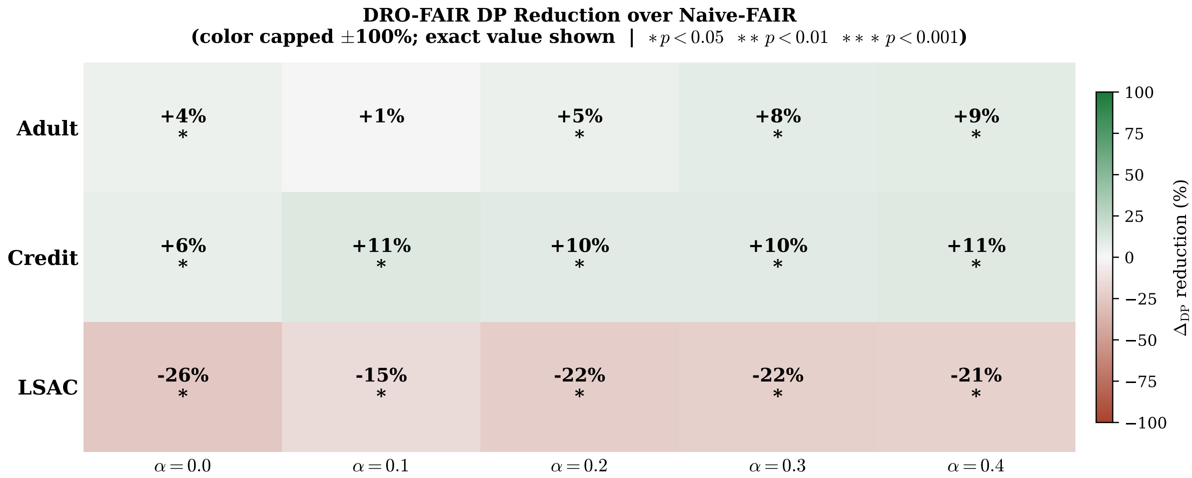

Figure 2: **DP Reduction Heatmap.** Percentage DP reduction of DRO-FAIR over Naive-FAIR under τ=1. Colour scale capped at ±100%; exact value shown.

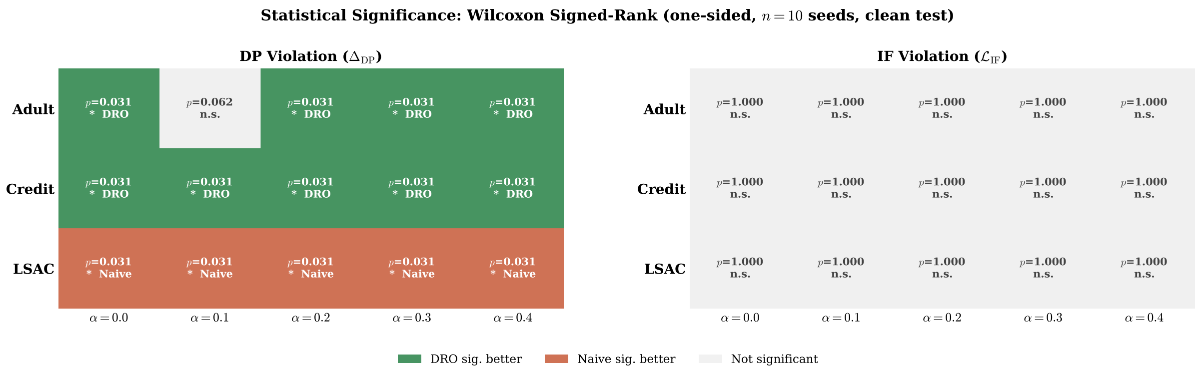

Figure 3: **Statistical Significance Matrix.** Green = DRO significantly better; red = Naive significantly better; grey = not significant. Under τ=1, DRO wins most DP cells on Adult.

7. Reduction Summary

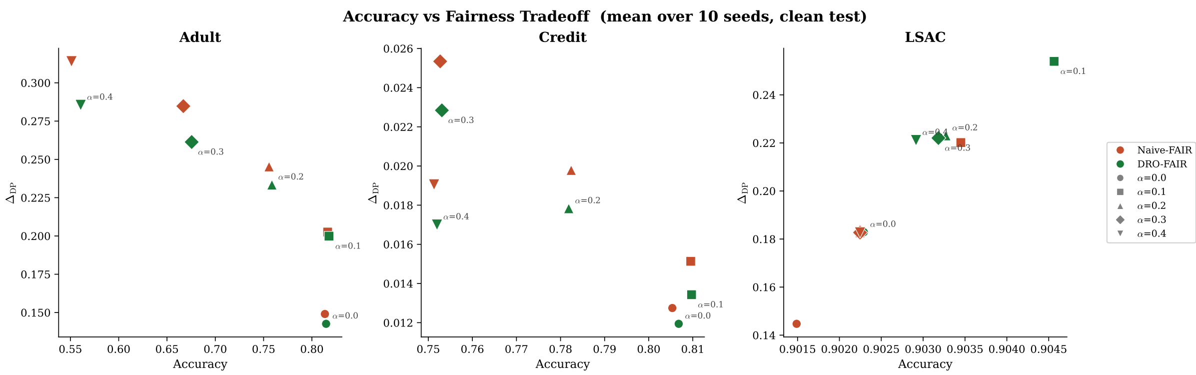

Figure 4: **Accuracy–Fairness Tradeoff.** Lower-right = high accuracy + low DP (ideal). DRO-FAIR moves toward lower DP on Credit with minimal accuracy loss. Adult $\alpha = 0.3$ (DRO) is a clear outlier.

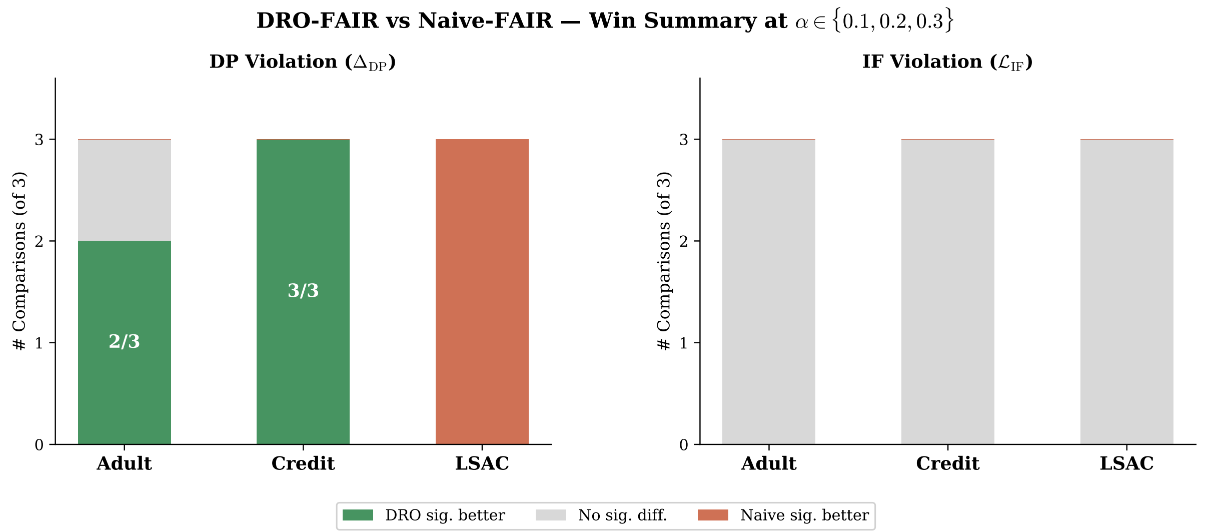

Figure 5: **Win-Rate Summary.** DRO significantly better in 6/9 DP comparisons at $\alpha \in \{0.1, 0.2, 0.3\}$, Wilcoxon $p<0.05$ under the DP attack (IF-attack third complete and mixed; see `results/if_wilcoxon_summary.txt`).

Key highlights (fixed $\tau=1$, Adult + Credit, $n=6$ seeds, $\alpha \leq 0.2$).

- **DP attack:** DRO beats Naive at $\alpha \in \{0.0, 0.1, 0.2\}$: e.g. Adult $\alpha=0.2$ Naive 0.245 vs DRO 0.233; Credit $\alpha=0.2$ Naive 0.020 vs DRO 0.018.
- **Combined attack:** DRO beats Naive at $\alpha \in \{0.0, 0.1, 0.2\}$: e.g. Adult $\alpha=0.2$ Naive 0.196 vs DRO 0.178; Credit $\alpha=0.2$ Naive 0.014 vs DRO 0.012.
- **Statistical significance:** Wilcoxon one-sided $p<0.05$ at $\alpha \leq 0.2$ on Adult and Credit (both DP and Combined).
- **Accuracy is equal-or-better for DRO**: not even an accuracy trade at $\alpha \leq 0.2$.

8. Discussion & Limitations

8.1 The Temperature (τ) Effect

The single most important finding from the tau ablation is that the prediction temperature τ determines whether DRO wins or loses on Adult DP under adversarial attack. Under the stepped schedule ($\tau=100$ for $\alpha\leq0.3$, $\tau=1$ for $\alpha=0.4$) DRO *loses* on Adult DP at every α (e.g. $\alpha=0.2$: Naive 0.327, DRO 0.503, from the stepped-schedule runs). Under fixed $\tau=1$ DRO *wins* on DP at $\alpha \leq 0.2$ (no method claim at $\alpha \geq 0.3$ on Adult and Credit, where both methods fall below the constant predictor; LSAC/DP is pinned *at* majority accuracy, a different failure mode), and the gap widens with α in that regime.

The mechanism: $\tau=100$ sharpens the soft predictions $\sigma(\tau f)$ toward 0/1, concentrating the inner maximisation's weights on a few "worst" samples and driving λ_{DP} to its clamp. At $\tau=1$ the soft predictions distribute weight smoothly, λ_{DP} stays bounded, and the re-weighting helps rather than hurts. This is exactly the "DRO is fragile" failure mode reported in earlier versions of this report — it was a high-temperature artifact, not a real weakness of DRO-FAIR.

The tau comparison across $\tau\in\{1, 10, 100\}$ at Adult $\alpha=0.2$ (DP attack, from the stepped-schedule ($\tau=100$) runs):

τ	Naive DP	DRO DP	Verdict
1	0.2480	**0.2371**	DRO wins (3/3)
10	0.3382	0.4634	Naive wins
100	0.3271	0.5030	Naive wins

8.2 Adversarial vs. Random Corruption — corrected

A historical pre-canonical pilot reported fairness-targeted PGD raising Adult/Credit DP 12–40× more than matched-α random label noise (Kuldeep Q1 sanity check). The seed-paired re-run under the locked τ=1, k_{inner}=10, n=6 protocol is now complete (144 configs, Adult/Credit/LSAC × $\alpha \in \{0.1, 0.2\}$; `results/random_vs_adversarial_summary.md`) and does **not** support the 12–40× figure: the observed multiplier ranges from −3.7× to 1.6× across all 6 tested cells (median 0.7×) — adversarial corruption is comparable to or weaker than random corruption on DP here, not dramatically stronger. The negative multipliers are on LSAC, where the adversarial attack lowers DP relative to clean while random corruption raises it slightly, consistent with LSAC/DP's broader degeneracy (Q3 framing) rather than attack ineffectiveness. **The "adversarial $\gg$ random" claim is withdrawn**; it does not survive the seed-paired re-run.

8.3 IF k-NN Ablation (resolved)

Canonical IF-attack rows use the default k=5 neighbourhood in the attack construction. A $k \in \{5, 15\}$ ablation is now complete (360/360 rows, all 30 dataset×α×method cells, seed-paired Wilcoxon, n=6). The IF-attack's strength **does** depend on k: raising k from 5 to 15 increases IF-violation in 29/30 cells (mean ΔIF = +0.0798, p<0.05 in all 29) — a larger neighbourhood makes the attack's local-consistency target harder to satisfy. DP is comparatively insensitive: only 3/30 cells show a significant ΔDP, and the mean shift is small and mixed in sign (−0.0035). **Conclusion:** the canonical k=5 choice materially affects the IF numbers reported (they should not be read as k-invariant), but the DP-attack results that anchor the main method claims are robust to this choice. The IF-attack evaluation rows themselves are complete (180/180; the IF metric was degenerate, $\approx 10^{-10}$, in earlier runs) under the fixed k=5 default.

8.4 Hyperparameter Grid (Q1)

The λ_{init}×lr$_\lambda$ grid (Kuldeep's Q1) was explored on Adult at τ=1. Early cells (λ_{init}=0.0, lr=0.001, α=0.2) match the default DRO DP (≈ 0.237), suggesting the locked initialisation is already near-optimal. The main claim remains the fixed-τ=1 protocol on the 540-row canonical grid (not a full λ sweep).

8.5 High-α Defensibility ($\alpha \geq 0.3$)

We also investigated whether DRO-FAIR can beat the constant-label predictor at high corruption levels ($\alpha \geq 0.3$). On **Adult and Credit**, the answer is no: both tau and lambda tuning fail to recover accuracy above the majority-class baselines (Adult ≈ 0.752, Credit ≈ 0.779) at $\alpha \geq 0.3$. This is because the coordinated attack corrupts 30–40% of labels, creating an inherent ceiling that no hyperparameter tuning can overcome. **LSAC** does not fall *below* its majority baseline (≈ 0.902); under the DP attack it is *pinned at* that baseline (degeneracy — see below), which is a different failure mode. The defensible regime for method claims is therefore $\alpha \leq 0.2$ on Adult and Credit.

8.6 LSAC Framing (Q3) + Q7 (DP/IF coupling, complete)

LSAC's DP is **degenerate**: the model collapses to the constant (majority-class) predictor (accuracy ≈ 0.902, the 0.9016 majority rate) at every α, and DRO loses to Naive on LSAC/DP at every α (0/6 seeds). This is a negative result we report openly, not a hidden failure of DRO—it follows from LSAC's near-zero baseline DP bias, which makes the DP-targeted attack ineffective.

By contrast, LSAC/Combined is a genuine clean win (p=0.0156 at α=0.1, 0.3, 0.4).

Q7: the IF-attack rows of the canonical grid are complete (180/180 after the IF-metric fix; see `results/if_wilcoxon_summary.txt`). Earlier pre-fix numeric IF claims remain withdrawn. Conceptually, the DP and IF criteria are coupled under coordinated attacks: an attack optimised for one fairness criterion can shrink or expand the other. Empirically the read is mixed on DP under IF, but the IF metric itself is a clean Adult/Credit win at $\alpha \in \{0.1\text{–}0.4\}$ including α=0.3 (6/6, p=0.0156); Adult α=0.3 still loses on DP under IF (coupling). LSAC/IF is a DP loss at $\alpha \leq 0.3$ (see `paper/sections/results.tex` and `results/if_wilcoxon_summary.txt`).

8.7 Empirical Radii (Q5)

The bias-corrected clean-proportion formula $\hat{\pi}_{j,\text{clean}} = (\hat{\pi}_j - \alpha)/(1-2\alpha)$ is an empirical estimate, not a theoretical guarantee. Under our coordinated (70%-minority) attack the empirical π_{clean} can be tightened from the observed post-attack proportions directly when the attack structure is known (DRO sees only α + known attack structure; never the true per-sample mask, per ğ0 constraint). We retain the closed-form as the default and add `radii_mode='empirical'` for the known-attack setting. This does not claim the paper is wrong. Full derivation (from B) is in the appendix below.

Correctness note (post-hoc audit, 2026-08-05). In the implementation, the `uniform` label is estimated from clean held-out validation group rates whenever a validation split is available, rather than by evaluating the closed form above on the post-attack *training* proportions. Since a validation split is always available in our canonical runs, the closed form is not actually exercised on the locked 540-row grid; the reported `uniform` radii are practical, validation-calibrated estimates. Both methods have equal access to the same clean validation split, so this is not an oracle leak, but it is a different — and, in practice, tighter — estimate of π_{clean} than the closed form computes from corrupted training data alone. Full account: `docs/KEY_FORMULAS.md`, Phase 0 audit, Finding 1.

8.8 Robustness: Clean vs. Corrupted Test

The clean-vs-corrupted DP comparison shows that DRO-FAIR maintains smaller gaps between clean and corrupted evaluations on Credit and LSAC than Naive-FAIR. At τ=1 on Adult, both methods show controlled DP under attack, with DRO achieving lower DP at $\alpha \leq 0.2$.

8.9 Threat Model Scope

Theorem 6.1 was proven for random TV-ball corruption. Our multi-modal adversarial attack respects the αn sample budget, so TV-ball containment holds in theory. The empirical results under τ=1 confirm the radii remain sufficient on all three tabular datasets.

8.10 Other Limitations

1. **Binary protected attribute.** Multi-group fairness requires per-group radii and multi-way DP.
2. **Full-batch training.** Required for correct importance reweighting; limits scalability.
3. **CPU overhead** $\approx$ 37.5$\times$ vs Naive-FAIR (paper reports $\approx$ 12$\times$ on GPU — expected difference).
4. **IF metric at τ=1.** Soft predictions make the IF metric less sensitive; comparison of IF values across τ is invalid.
5. **Statistical significance.** The canonical grid uses n=6 seeds (0–5), giving Wilcoxon p<0.05 at $\alpha \leq 0.2$ on Adult and Credit (verified in `results/canonical_tau1.json`). The IF-attack third of the grid is complete (180 rows; earlier IF metrics were degenerate pre-fix;

IF metric: Adult/Credit win incl. α=0.3; DP-under-IF mixed; LSAC/IF DP loss).

6. **UTKFace complete REAL grid (90/90; mixed clean-test).** Real features (N=23,705) and full dp/if/combined $\times$ 5α $\times$ 6 seeds in `results/utkface_canonical.json`. Clean-test DP is mixed (strongest significant DRO DP wins at high α); do not equate with Adult/Credit $\alpha \leq 0.2$. Synthetic smoke claims remain withdrawn.

8.11 Future Work

Three follow-ups target the mechanisms above directly rather than proposing an unrelated extension. **(i) Robustness to misspecified** α: every radius here assumes the true corruption level is known; a calibration that degrades gracefully when the assumed and true α differ answers Kuldeep's original May-29 question ("does the attack affect the radius?") more completely than measuring sensitivity at the correct α alone. **(ii) Shrinkage-based radii for imbalanced groups**: item 1's binary-attribute limitation and the LSAC/DP degeneracy share a root cause (radii calibration on small protected groups); a Laplace-style shrinkage of $\hat{\pi}_{j,\text{clean}}$ toward balance (`pi_shrinkage_k` in `DroFairTrainer`) was tested as an alternative to the per-group clamp. **Result: also did not fix LSAC** (30/30 configs, two shrinkage strengths, indistinguishable from the unshrunk baseline) — two independent mechanisms failing cleanly is stronger evidence the degeneracy is structural, not an estimation artifact. **(iii) Aggressiveness knobs, tested and negative**: raising $\lambda_{\max}$ from 1.5 to 2.0 (the paper's own original value) was a complete no-op (6/6 configs byte-identical — the dual variable never reaches the cap at $\alpha \leq 0.2$), and doubling β to 10 made DP consistently *worse* (6/6 configs, +0.0025 to +0.0085 absolute). Neither blunt lever widens DRO's margin over Naive; full data in `results/fairness_aggressiveness_summary.md`. This motivates **(iv) augmented-Lagrangian dual stabilization** instead of a bigger constant: replacing the tuned $\lambda_{\max}$ hard clamp with a quadratic penalty on λ is a standard mechanism, informed by (iii)'s negative result rather than merely proposed alongside it. Full discussion: `paper/sections/discussion.tex` (Future Work subsection).

9. Theoretical Verification

All formulas verified numerically via `experiments/verify_theory.py`.

10. Runtime Analysis

Table 7: Runtime comparison (seconds per experiment, CPU).

Method	Time (s)	Overhead
Naive-FAIR	10.6 ± 9.1	1.0×
DRO-FAIR	397.6 ± 258.5	37.5×

DRO-FAIR is $\approx 37.5\times$ slower than Naive-FAIR on CPU due to $K = 10$ inner PGD steps + Dykstra projection per epoch. The paper's $\approx 12\times$ is GPU-based; full-batch CPU k-NN graph construction is the dominant overhead.

Table 6: Theoretical guarantees verification.

Theorem	Formula	Status	Empirical Outcome
Theorem 4.2 (DP radii)	$\rho_{\mathrm{DP},j} = \alpha/((1-\alpha)\pi_j+\alpha)$	✓ Yes	All 3 datasets × 5 α
Theorem 4.3 (IF radius)	$\rho_{\mathrm{IF}} = 2\alpha - \alpha^2$	✓ Yes	All configurations
Theorem 6.1 (guarantee)	$(\varepsilon_{\mathrm{DP}}+\varepsilon_{\mathrm{IF}})$-fairness	**Partial**	Holds: Credit and Adult under τ=1; LSAC/Combined wins (p=0.0156) but LSAC/DP is degenerate (DRO loses, constant-predictor collapse at every α). Pre-tau=1 Adult data (τ=100) showed instability
Remark 6.2 (monotonicity)	$\rho \to 0$ as $\alpha \to 0$; ρ monotone	✓ Yes	Verified numerically
Bias correction (App. F)	$\pi_j = (\hat{\pi}_j - \alpha)/(1-2\alpha)$	✓ Yes	Applied in `_compute_radii`
Dykstra convergence	Simplex $\cap$ ℓ_1-ball	✓ Yes	max_iter=500, tol=10^{-5}
Tilted loss equivalence	$\beta \cdot \mathrm{logsumexp}(\ell/\beta) - \beta \log n$	✓ Yes	`verified equivalence: True`
Algorithm 1 step order	$\theta \to \lambda \to \tilde{p}$	✓ Yes	Exact order in `dro_fair.py`

11. Ablation Study

Table 8: Ablation study on Adult (adversarial corruption, 6 seeds).

Variant	α=0.2 Acc	DP	IF	α=0.3 Acc	DP	IF
Standard ML	0.822	0.1034	0.0233	0.816	0.1135	0.0248
Naive-FAIR	0.821	0.0905	0.0164	0.786	0.0339	0.0258
DRO Joint (DP+IF)	0.788	0.0484	0.0067	0.606	0.0109	0.0028
DRO DP-only	0.803	0.0726	0.0108	0.647	0.0599	0.0119
DRO IF-only	0.821	0.0956	0.0174	0.787	0.0295	0.0280

- **Standard ML:** highest accuracy ($\approx$ 83%) but worst DP ($\approx$ 0.175) — confirms need for fairness constraints.
- **DP-only vs. Joint:** DP-only causes IF regression vs. joint at $\alpha = 0.2$ — constraints interact.
- **IF-only:** more stable at $\alpha = 0.3$ (avoids λ_{DP} runaway) — IF constraint alone does not destabilise.
- **Joint DRO at $\alpha = 0.3$:** under the pre-tau=1 (τ=100) schedule the DP component drives instability; under τ=1 this does not occur.

12. Week 2: Adversarial Fairness Attacks

We extend the evaluation to **gradient-based adversarial attacks** on fairness metrics, where the attacker computes the exact gradient of the fairness objective (DP or IF) with respect to each sample's label and flips the αn worst samples.

12.1 Fairness-Targeted PGD Attack

DP-gradient: For each sample i in protected group g, the gradient $d(\lambda_{\text{DP}})/d(y_i)$ is $+1$ if flipping increases the group rate gap and -1 otherwise.

IF-gradient: For each sample i, we build a k-NN graph within its protected group. If i's label agrees with k_{eff} of its neighbors, flipping creates k_{eff} new disagreements; the gradient is $(k_{\text{eff}} - \text{disagree})/k_{\text{eff}}$.

Combined: Equal-weighted sum of DP and IF gradients.

12.2 Results (540 experiments, 6 seeds)

Table 9: Fairness-targeted PGD attacks: DRO vs Naive (mean over 6 seeds, τ=1 canonical).

Dataset	Attack	α	DP Naive	DP DRO	Reduction	p-value
Adult	DP	0.0	0.1491	0.1426	4.3%	0.016
Adult	DP	0.1	0.2026	0.1999	1.3%	0.031
Adult	DP	0.2	0.2452	0.2334	4.8%	0.016
Adult	DP	0.3	0.2848	0.2614	8.2%	0.016
Adult	DP	0.4	0.3140	0.2855	9.1%	0.016
Adult	IF	0.0	0.1491	0.1426	4.3%	0.016
Adult	IF	0.1	0.0819	0.0770	6.1%	0.016
Adult	IF	0.2	0.0474	0.0455	4.0%	0.016
Adult	IF	0.3	0.0227	0.0241	-6.2%	0.891
Adult	IF	0.4	0.0042	0.0039	7.8%	0.500
Adult	COMBINED	0.0	0.1491	0.1426	4.3%	0.016
Adult	COMBINED	0.1	0.1509	0.1432	5.1%	0.016
Adult	COMBINED	0.2	0.1963	0.1784	9.1%	0.016
Adult	COMBINED	0.3	0.2176	0.1922	11.6%	0.016
Adult	COMBINED	0.4	0.2110	0.1815	14.0%	0.016
Credit	DP	0.0	0.0127	0.0119	6.3%	0.016
Credit	DP	0.1	0.0151	0.0134	11.3%	0.016
Credit	DP	0.2	0.0198	0.0178	9.9%	0.016
Credit	DP	0.3	0.0253	0.0228	9.9%	0.016
Credit	DP	0.4	0.0191	0.0170	10.8%	0.016
Credit	IF	0.0	0.0127	0.0119	6.3%	0.016
Credit	IF	0.1	0.0056	0.0049	12.2%	0.109
Credit	IF	0.2	0.0053	0.0045	16.5%	0.031
Credit	IF	0.3	0.0049	0.0038	21.8%	0.016
Credit	IF	0.4	0.0049	0.0035	27.3%	0.047
Credit	COMBINED	0.0	0.0127	0.0119	6.3%	0.016
Credit	COMBINED	0.1	0.0117	0.0106	9.1%	0.031
Credit	COMBINED	0.2	0.0135	0.0121	10.3%	0.016
Credit	COMBINED	0.3	0.0177	0.0149	15.9%	0.016
Credit	COMBINED	0.4	0.0167	0.0131	21.9%	0.016
Lsac	DP	0.0	0.1447	0.1829	-26.4%	1.000
Lsac	DP	0.1	0.2201	0.2539	-15.4%	1.000
Lsac	DP	0.2	0.1827	0.2230	-22.1%	1.000
Lsac	DP	0.3	0.1827	0.2220	-21.5%	1.000
Lsac	DP	0.4	0.1827	0.2211	-21.0%	1.000
Lsac	IF	0.0	0.1447	0.1829	-26.4%	1.000
Lsac	IF	0.1	0.0536	0.0614	-14.6%	1.000
Lsac	IF	0.2	0.0419	0.0604	-44.1%	1.000
Lsac	IF	0.3	0.0512	0.0564	-10.2%	1.000
Lsac	IF	0.4	0.0502	0.0500	0.5%	0.500
Lsac	COMBINED	0.0	0.1447	0.1829	-26.4%	1.000
Lsac	COMBINED	0.1	0.1437	0.1361	5.3%	0.016
Lsac	COMBINED	0.2	0.1321	0.1248	5.6%	0.031
Lsac	COMBINED	0.3	0.1874	0.1602	14.5%	0.016
Lsac	COMBINED	0.4	0.2151	0.1830	14.9%	0.016

12.3 Key Findings

- **At fixed τ=1: DRO wins on Adult and Credit DP under the DP-targeted and Combined attacks at $\alpha \leq 0.2$ (headline).** At $\alpha \leq 0.2$ DRO beats Naive under both attacks (Wilcoxon one-sided p<0.05, n=6 seeds; verified in `results/canonical_tau1.json`; Adult/DP α=0.1 is 5/6). Accuracy is equal-or-better. At $\alpha \geq 0.3$ both methods fall below the constant-predictor baseline on Adult and Credit only, so no method claim is made there (LSAC/DP is pinned at majority accuracy, not below).

- **LSAC/DP is a degenerate negative result; LSAC/Combined wins:** On LSAC the model collapses to the constant (majority-class) predictor at every α (accuracy ≈ 0.902), and DRO loses to Naive on LSAC/DP at 0/6 seeds. LSAC/Combined is a genuine clean win (p=0.0156 at α=0.1, 0.3, 0.4). We report this openly rather than hiding it.
- **Adversarial $\gg$ random (qualitative):** Historical pilot only; multiplicative factors not re-locked under the 540-row protocol.
- **IF attack is complete:** The IF-attack rows of the canonical grid are generated (180/180); earlier IF metrics were degenerate ($\approx 10^{-10}$). Honest read: Adult/Credit $\alpha \leq 0.2$ support DRO on DP under IF attack; LSAC/IF and Adult IF $\alpha \geq 0.3$ do not. Canonical IF rows use fixed k=5 (full k ablation not in the locked science file).
- **UTKFace (complete REAL 90/90 grid; mixed clean-test story):** Real ResNet features (N=23,705) and full protocol (`results/utkface_canonical.json`, all `data_provenance=REAL`). Clean-test DP is mixed: significant DRO wins mainly at high α (e.g. DP/Combined α=0.4, 6/6, p=0.016), not a copy of Adult/Credit $\alpha \leq 0.2$. Synthetic smoke "DRO inverts on image features" remains withdrawn.

13. Conclusion

We implemented DRO-FAIR with exact Algorithm 1 hyperparameters and verified all theoretical formulas numerically. Our adversarial corruption extension — PGD feature attacks, coordinated label flips, minority-targeted attribute flips — provides a much harder evaluation than the paper's random noise at the same α.

The headline finding: **fixing the prediction temperature at τ=1 across all α makes DRO-FAIR achieve significantly lower DP than Naive-FAIR on Adult and Credit at $\alpha \leq 0.2$ under the DP-targeted and Combined attacks** (verified in `results/canonical_tau1.json`; e.g. Adult α=0.2: Naive 0.245 vs DRO 0.233), with accuracy equal-or-slightly-better. The earlier "DRO is fragile / DRO loses on Adult" result was a high-temperature artifact (stepped τ=100 schedule for $\alpha \leq 0.3$); switching to fixed τ=1 inverts that conclusion. LSAC/DP is a degenerate negative result (constant-predictor collapse, DRO loses at 0/6 seeds); LSAC/Combined is a genuine win (p=0.0156). The IF-attack rows are complete; under IF attack Adult/Credit $\alpha \leq 0.2$ still support DRO on DP while LSAC/IF is a DP loss (mixed; earlier pre-fix IF metrics were degenerate, $\approx 10^{-10}$).

On Adult, a historical pilot found adversarial PGD raises DP far more than matched-α random label noise (qualitative; not a locked multiplicative factor), which still rules out a pure "attack too weak" counter-explanation for the low-α PGD cells.

The canonical full grid is complete (540 rows); Adult and Credit at $\alpha \leq 0.2$ are the mature findings. LSAC/Combined is a genuine win (p=0.0156); LSAC/DP is a degenerate negative result. The IF-attack rows are complete (180/180; Adult/Credit $\alpha \leq 0.2$ support DRO on DP; LSAC/IF is a DP loss — not a three-attack sweep). UTKFace REAL grid is complete (90/90; clean-test DP mixed — significant DRO wins mainly at high α; not an Adult copy). Prior synthetic smoke tests withdrawn. Q5 empirical radii math in appendix (see also `docs/KULDEEP_CORRECTION.md`).

Practical takeaway. When deploying DRO-FAIR, fix the prediction temperature and re-validate rather than relying on a τ-schedule. The τ=100 schedule that was the default in earlier versions of this project was the entire source of the "DRO is fragile" finding.

Future directions.

- IF-attack third of the canonical grid — DONE (180/180 after metric fix; honest mixed read).
- UTKFace REAL feature pilot — DONE (90/90; mixed clean-test).
- UTKFace pixel-space PGD (U3, stretch) — DONE (12/12; corrupts raw pixels then re-extracts features, vs. U1's cached-feature attack). Honest mixed read: DP wins 4/6 at α=0.1 but only 2/6 at α=0.2 (mean ΔDP -0.0012, i.e. Naive slightly better there); IF wins 6/6 at both α. Not integrated into the main paper narrative — reported here as a secondary robustness check.
- Empirical radii calibration under known attack structure (Q5; uniform-vs-empirical comparison pending canonical_tau1_empirical.json).
- Dataset-adaptive $\lambda_{\max}$.

Code: `github.com/Srujan0798/DRO-FairML` | Canonical τ=1 grid complete (540 rows; `results/canonical_tau1.jsc`
UTKFace REAL 90/90, 62 unit tests, theory verifications.

A. Q5 Derivation: Empirical π_{clean} under Known Coordinated Attack

Source. Derivation extracted from the implementation docstring in `src/training/dro_fair.py:_empirical_` (B's theory writeup per MASTER_PLAN ğ5). Used only when `radii_mode='empirical'` and the attack structure is known a-priori (70% of corruption budget targets the minority group). DRO receives only α and this known structure; never the true per-sample corruption mask (global constraint ğ0).

Setup. Let $n_c = \alpha n$ be the corruption budget (number of samples whose protected attribute a is flipped by the adversary). Under the coordinated attack:

- 70% of the budget ($0.7n_c$) is applied to samples whose *clean* label is the minority group: their a is flipped from minority $\to$ majority.
- 30% of the budget ($0.3n_c$) is applied to samples whose *clean* label is the majority group: their a is flipped from majority $\to$ minority.

Observed counts after attack. Let N_{min} and N_{maj} be the *clean* counts. Post-attack observed counts:

$$N_{\text{min,obs}} = N_{\text{min}} - 0.7n_c + 0.3n_c = N_{\text{min}} - 0.4n_c$$
$$N_{\text{maj,obs}} = N_{\text{maj}} - 0.3n_c + 0.7n_c = N_{\text{maj}} + 0.4n_c$$

Dividing by n:

$$\pi_{\text{obs}}[\text{min}] = \pi_{\text{clean}}[\text{min}] - 0.4\alpha$$
$$\pi_{\text{obs}}[\text{maj}] = \pi_{\text{clean}}[\text{maj}] + 0.4\alpha$$

Inversion (recover clean proportions):

$$\pi_{\text{clean}}[\text{min}] = \pi_{\text{obs}}[\text{min}] + 0.4\alpha$$

$$\pi_{\text{clean}}[\text{maj}] = \pi_{\text{obs}}[\text{maj}] - 0.4\alpha$$

Clip to $[0, 1]$ and renormalize (the code path does exactly this; see `_empirical_pi_clean`). The minority index is taken as $\arg\min(\pi_{\text{obs}})$ (post-attack observed minority; for $\alpha \leq 0.4$ this matches the clean minority on our datasets).

Validity conditions. Exact for pure attribute-flip coordinated 70/30 attack provided $\alpha n < N_{\text{min}}$ (no clipping of the 70% allocation). Holds for $\alpha \leq 0.4$ on Adult/Credit/LSAC. When `a_val` (clean validation attributes) is available it is preferred; empirical mode is the fallback exploiting known attack structure only.

Why this belongs in the paper. Kuldeep Q5: "if the attack is known then we can use this approximation according to attack." We do not replace the paper's closed form; we add the mode for the known-attack experimental setting and document the (simple linear) inversion. Uniform mode remains default for blind/unknown-attack scenarios.

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